Chapter 03

- reversed Euler's formula

$e^{j\omega t} = \cos(\omega t) + j\sin(\omega t)$

- $\cos(\omega t) = \frac{1}{2}(e^{j\omega t} + e^{-j\omega t})$

$e^{-j\omega t} = \cos(\omega t) - j\sin(\omega t)$

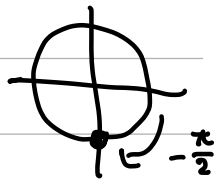
- $\sin(\omega t) = \frac{1}{2j}(e^{j\omega t} - e^{-j\omega t})$

- 예제

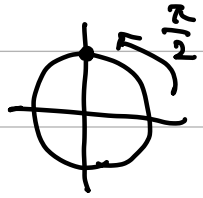
$A\sin(\eta t) = \frac{A}{2j}e^{j\eta t} - \frac{A}{2j}e^{-j\eta t}$ * A > 0 가정 유의!

$\frac{A}{2j} = \frac{A}{2}\cdot\frac{1}{j} = \frac{A}{2}\cdot(-j) = \frac{A}{2}\cdot e^{-j\frac{\pi}{2}}$

$(\frac{1}{j} = \frac{1}{j}\cdot\frac{j}{j} = \frac{j}{-1} = -j)$, $(-j = e^{-j\frac{\pi}{2}})$



$-\frac{A}{2j} = \frac{A}{2}(-\frac{1}{j}) = \frac{A}{2}\cdot j = \frac{A}{2}\cdot e^{j\frac{\pi}{2}}$



$\therefore A\sin(\eta t) = \frac{1}{2}Ae^{-j\frac{\pi}{2}}e^{j\eta t} + \frac{A}{2}e^{j\frac{\pi}{2}}e^{-j\eta t}$

- DC (Direct Current) → constant, zero-frequency component

  represent Avg. value

  always $\varphi = 0$ or $-\pi$ → π?

- inverse Euler's formula

$$A\cos(\omega t + \varphi) = \frac{1}{2} A e^{j\varphi} e^{j\omega t} + \frac{1}{2} A e^{-j\varphi} e^{-j\omega t}$$

- general form

$$x(t) = A_0 + \sum_{k=1}^{N} A_k \cos(2\pi f_k t + \varphi_k)$$

$$= X_0 + \sum_{k=1}^{N} \mathrm{Re}\left\{ X_k e^{j 2\pi f_k t} \right\}$$

$X_0 \rightarrow A_0 e^{j0}$

$X_k = A_k e^{j\varphi_k}$

* $\mathrm{Re}\{z\} = \frac{1}{2} z + \frac{1}{2} z^*$

$$= X_0 + \sum_{k=1}^{N} \left\{ \frac{1}{2} X_k e^{j2\pi f_k t} + \frac{1}{2} X_k^* e^{-j2\pi f_k t} \right\}$$

$z = a + jb$

$z^* = a - jb$

$z + z^* = 2a$

- Periodic waveforms

$f_k = k f_0$

↓

int

- $\cos(3t)$ $\omega = 3$, $T = \frac{2\pi}{\omega} = \frac{2\pi}{3}$

↓ 제곱

$x(t) = \cos^2(3t) = \frac{1}{2}(\cos 6t + 1)$ $\omega = 6$, $T = \frac{2\pi}{6} = \frac{\pi}{3}$

- $x(t+T) = x(t)$ if period is T

$e^{j\omega(t+T)} = e^{j\omega t}$ $f_0 = \frac{1}{T}$

$e^{j\omega T} = 1$

$\omega T = 2\pi k$ $\therefore \omega = \frac{2\pi k}{T} = \underbrace{\left(\frac{2\pi}{T}\right)}_{2\pi f_0} k = \underbrace{\omega_0}_{\text{fundamental radian frequency}} k$ (k는 정수)

- $f_0 = GCD(200, 400, 500, 1600, 1700) = 100$

- Fourier Series

synthesis vs analysis

$x(t)$ 생성 $x(t)$에서 추출 $\rightarrow a_k$